From Euclidean Lattice to Light Cone:

Essential Papers on Lattice QCD Parton Physics

An annotated selection of 38 landmark papers
distilled from a 99-paper research library and its citation network

Preface: How This Selection Was Made

This booklet distills a working research library of **99 lattice-QCD articles** (the `papers/` collection) into the set of papers that matter most. The selection draws on two and only two sources, as required:

1. the **original articles themselves**, several of which are classics of the field in their own right; and
2. their **reference lists**: every bibliography of every article was mined, and the citation counts were tallied across the whole corpus.

A paper earns its place here for one (or both) of two reasons: (i) it is an *article of the library itself* whose content is a milestone of the field, or (ii) it is a *hub of the citation network* — a foundational work that the library articles lean on again and again. The citation statistics quoted below are conservative estimates from this particular corpus; they measure relevance *to this library*, not global impact (which would only be larger).

The map of the territory

The 38 selected papers organize themselves naturally into eight chapters, which are also the eight load-bearing pillars of modern lattice parton physics:

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1	QCD and the parton idea: theoretical foundations	3
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Two storylines run through the whole book and meet in Chapter 6:

- the **gradient-flow line** (Lüscher's program): smooth gauge fields as renormalized fields, from trivializing maps to smeared quasi-distributions;
- the **large-momentum line** (Ji's LaMET and Radyushkin's pseudo-PDFs): boosted equal-time correlators matched perturbatively onto light-cone distributions.

Around them orbit the enabling technologies (Chapter 2), the hard renormalization problems that made first-principles PDFs possible at all (Chapter 5), and the two interfaces through which lattice results reach the rest of physics: global PDF fits (Chapter 7) and machine-learning samplers rooted in stochastic quantization (Chapter 8).

How to read the entries

Each entry opens with a bibliographic card, then answers three questions: *why* it was selected (with its hub status inside this corpus), *what* it actually says (physics, with key formulas), and where its *echoes* appear in the library. Entries are self-contained; formulas are reproduced in modern notation for orientation, not as facsimiles.

A note on honesty. Citation counts below are counts within this 99-article corpus ("∼N/99"), rounded conservatively. Bibliographic data were transcribed from the reference lists of the library articles themselves and cross-checked against standard records; any field that could not be confirmed is marked [?].

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Chapter 1

QCD and the Parton Idea: Theoretical Foundations

Everything in this book rests on three discoveries of the early 1970s: that the strong interaction becomes weak at short distances, that parton distributions obey renormalization-group evolution, and that the theory can be regularized on a spacetime lattice without losing gauge invariance. The library cites all three whenever it needs to say why lattice QCD is worth the effort.

1.1 Asymptotic Freedom

Ultraviolet behavior of non-abelian gauge theories
D. J. Gross and F. Wilczek
Phys. Rev. Lett. **30**, 1343 (1973); companion paper: H. D. Politzer, *Reliable perturbative results for strong interactions?*, Phys. Rev. Lett. **30**, 1346 (1973)
No arXiv (preprint era) | DOI: 10.1103/PhysRevLett.30.1343 / .1346

Why selected. The discovery that non-abelian gauge theory is asymptotically free is the single fact that makes QCD a calculable theory and parton distributions meaningful objects. It is cited in the library wherever the reason for doing lattice QCD is stated (e.g. the BMW spectrum and neutron–proton mass-difference papers, and the lattice-QCD thesis in the collection).

Physics content. In Yang–Mills theory with n_f quark flavors the one-loop beta function is negative,

$$\beta(g) = -\frac{g^3}{16\pi^2}\Big(11 - \frac{2}{3}n_f\Big) + \mathcal{O}(g^5), \tag{1.1}$$

so the coupling decreases logarithmically at short distance, $\alpha_s(Q^2) \to 0$ as $Q^2 \to \infty$. Deep-inelastic scattering then sees nearly free quarks (Bjorken scaling to leading order), while at large distances the coupling grows — confinement is not proven here but made plausible as the same phenomenon viewed from the infrared side. The sign of β_0 is fixed by the gluon self-interaction: only non-abelian gauge theories screen antiscreened.

Impact and echoes in this library. Every global PDF analysis in the library (CT18, NNPDF, CJ, JAM) evolves its partons with splitting functions whose existence is guaranteed by this result; every lattice paper invokes it when motivating QCD as the theory to simulate. It is the root of the entire tree.

1.2 DGLAP Evolution

Asymptotic freedom in parton language
G. Altarelli and G. Parisi
Nucl. Phys. B **126**, 298 (1977)
No arXiv (preprint era)

Why selected. The DGLAP equation translated the operator-product-expansion view of scaling violations into the parton language in which every PDF paper of this library is written. It appears in the reference lists of the Ji 2013 founding paper, the pseudo-PDF papers of Radyushkin and Orginos et al., the gradient-flow PDF papers, and many more (∼8/99).

Physics content. Quark distributions are not constants but scale-dependent amplitudes obeying an evolution equation of probabilistic form,

$$\frac{\partial q(x,\mu)}{\partial \ln \mu^2} = \frac{\alpha_s(\mu)}{2\pi}\,(P_{qq}\otimes q + P_{qg}\otimes g), \qquad (P\otimes q)(x) = \int_x^1 \frac{dz}{z}\,P(z)\,q\Big(\frac{x}{z},\mu\Big)\,. \tag{1.2}$$

The splitting functions P_{ij} are computed from collinear factorization; in Mellin moment space they become matrices acting on moments $M_n = \int dx\, x^{n-1} q(x)$, which is exactly the language of lattice moments calculations. DGLAP is the "vertical" axis of every plot in this book's subject area: any distribution extracted on the lattice at some scale μ must be evolved before comparison with phenomenology.

Impact and echoes in this library. The matching kernels of quasi- and pseudo-PDFs reproduce the one-loop P_{ij} in their small-z limit — this consistency check appears verbatim in Ji 2013, Radyushkin 2017, Balitsky–Morris–Radyushkin 2020 and the smeared-quasi-PDF papers. Modern three-loop splitting functions (Moch–Vermaseren–Vogt 2004) underpin the NNLO fits used as benchmarks throughout the library.

1.3 Lattice Gauge Theory

Confinement of quarks
Kenneth G. Wilson
Phys. Rev. D **10**, 2445–2459 (1974)
No arXiv (preprint era) | DOI: 10.1103/PhysRevD.10.2445

Why selected. The founding document of lattice field theory and the conceptual ancestor of every calculation in the library. Within this corpus it is cited by the BMW mass papers, the AI-sampling papers, the agentic-computing papers, and the chiral-phase-transition thesis (∼4/99 directly; effectively universal). A full English LaTeX transcription of this paper exists in the companion `books/` collection together with its Chinese translation.

Physics content. Wilson replaced continuum spacetime by a hypercubic lattice while keeping gauge invariance exact: the degrees of freedom are link variables $U_\mu(n) \in SU(3)$, parallel transports along links, and the action is built from plaquettes,

$$S_W = \frac{\beta}{3} \sum_{n,\mu<\nu} \Big(1 - \frac{1}{3}\mathrm{Re}\,\mathrm{Tr}\, U_{\mu\nu}(n)\Big), \qquad \beta = \frac{2N_c}{g^2}. \tag{1.3}$$

Gauge invariant observables are traces of products of links along closed paths — Wilson loops. The area law $\langle W(C)\rangle \sim e^{-\sigma A}$ signals confinement through a nonzero string tension σ, and the static quark potential rises linearly. Strong-coupling expansion proves the area law at $\beta = 0$; asymptotic freedom guarantees the Coulomb law at $\beta \to \infty$: both regimes live on one lattice, connected by the crossover that Monte Carlo (Chapter 2) was invented to explore.

Impact and echoes in this library. Every ensemble, every correlator, every flow time in the library's 99 papers exists because Wilson wrote this paper. His name also marks the Wilson line — the path-ordered exponential of links — which will turn out to be precisely the object whose renormalization (Chapters 5–6) decides whether parton physics can be extracted from the lattice at all.

Chapter 2

Simulation Engines and Signal Enhancement

Between Wilson's action and a physics number stands machinery: an algorithm that samples gauge fields, an improved fermion discretization, and smearing techniques that make hadron signals visible at the large momenta demanded by parton physics. The four papers of this chapter are the load-bearing members of that machinery; two more (the Wuppertal smearing of Güsken et al. 1989 and the clover term of Sheikholeslami–Wohlert 1985) are used so pervasively by the library's ensembles that they are acknowledged here without separate entries.

2.1 Hybrid Monte Carlo

Hybrid Monte Carlo
S. Duane, A. D. Kennedy, B. J. Pendleton, D. Roweth
Phys. Lett. B **195**, 216–222 (1987)
No arXiv (preprint era) | DOI: 10.1016/0370-2693(87)90011-7

Why selected. HMC is the engine behind essentially every dynamical ensemble used in this library (MILC HISQ, CLS, clover ensembles of CLQCD, domain wall). It is cited explicitly by the HMC auto-tuning paper found in the collection, by both flow-based-sampling papers, and by the charmed-meson ensemble paper.

Physics content. HMC marries molecular dynamics to Metropolis. Momenta p are attached to the field variables; the system is integrated along a fictitious Hamiltonian trajectory of length τ,

$$H = \frac{1}{2}p^2 + S[U], \qquad \dot{U} = i\left[\frac{\partial H}{\partial p}\right]U, \tag{2.1}$$

and the endpoint is accepted with probability $\min(1, e^{-\Delta H})$. Because the integrator is reversible and area-preserving, detailed balance is exact while the global proposal suppresses random-walk behavior: decorrelation grows like τ rather than like step count. The cost — repeated Dirac inversions — drove thirty years of solver innovation, culminating in the multigrid and mixed-precision algorithms on which libraries such as QUDA (used by this corpus) are built.

Impact and echoes in this library. Every ensemble in every lattice PDF calculation of this library was generated by HMC or a direct descendant (Hasenbusch mass preconditioning,

multiple time scales). The AI-sampling line of Chapter 8 defines itself as an alternative proposal layer on top of the same Metropolis logic — and cites this paper as its baseline.

2.2 Analytic Link Smearing

Analytic smearing of $SU(3)$ link variables in lattice QCD
C. Morningstar and M. Peardon
Phys. Rev. D **69**, 054501 (2004)
arXiv:hep-lat/0307022 | DOI: 10.1103/PhysRevD.69.054501

Why selected. Stout smearing is cited throughout the gradient-flow chapters of this library as the discrete ancestor of the flow, and appears in the reference lists of Lüscher's own papers (∼7/99 among the flow corpus). It is also the standard noise-reduction tool in the smeared-quasi-PDF calculations.

Physics content. Each thin link is replaced by a smoothed link obtained by one exponential step toward the staple force,

$$U_\mu(n) \;\to\; e^{\,\mathrm{i}\rho\,\Omega_\mu(n)}\,U_\mu(n), \qquad \Omega_\mu = \mathrm{antiherm}\Big(\sum_{\pm\nu\neq\mu} U_\nu(n)U_\mu(n+\hat{\nu})U_\nu^\dagger(n+\hat{\mu})\Big), \tag{2.2}$$

iterated a few times. The update is analytic in the links, gauge covariant by construction, and cheap. The connection to the gradient flow is intimate: stout smearing with weight ρ is precisely the Euler discretization of the flow equation (3.2) of Chapter 3 with flow time $t = \rho$ per step.

Impact and echoes in this library. The equivalence "stout = coarse-grained flow" is how Lüscher 2010 motivates the physical meaning of the flow; Monahan and Orginos exploit the same structure when they smear quasi-PDF operators with the continuous flow (Chapter 6). Within the library it is also the workhorse smearing for nucleon-structure matrix elements.

2.3 Distillation

A novel quark-field creation operator construction for hadronic physics in lattice QCD
M. Peardon, J. Bulava, J. Foley, C. Morningstar, J. Juge, A. Jüttner, K. Orginos, et al. (Hadron Spectrum Collaboration)
Phys. Rev. D **80**, 054506 (2009)
arXiv:0905.2160 | DOI: 10.1103/PhysRevD.80.054506

Why selected. An original article of this library, and the enabling technology of the entire HadStruc pseudo-PDF program contained in it (gluon PDFs, gluon helicity, high-momentum distillation): cited by at least 4 articles of the corpus as method source.

Physics content. Instead of Jacobi smearing applied ad hoc at source or sink, distillation fixes once and for all a low-mode projector built from the three-dimensional Laplacian on each time slice,

$$\Box^{[N]}(t) = \sum_{k=1}^{N} |v_k(t)\rangle\langle v_k(t)|, \qquad -\Box_3\, v_k = \lambda_k v_k, \tag{2.3}$$

and writes every quark propagator through the *perambulator* τ, a small object computed once per configuration: $\langle x|e^{-Ht}|y\rangle = \Phi(x,t)\,\tau(t)\,\Phi^\dagger(y,0)$. Any number of interpolators can then be combined into variational matrices *post hoc*, with no new Dirac solves. The variational method of Lüscher–Wolff becomes affordable at scale.

Impact and echoes in this library. In this library, Egerer et al.'s *Distillation at High Momentum* extends the idea with momentum phases to reach nucleon momenta above 3 GeV — exactly the regime where the pseudo-ITD gluon calculations (Khan et al., Egerer et al.) operate. The HadStruc gluon-PDF line is inconceivable without this paper.

2.4 Momentum Smearing

Novel quark smearing for hadrons with high momenta in lattice QCD
G. S. Bali, B. Lang, B. U. Musch, A. Schäfer
Phys. Rev. D **93**, 094515 (2016)
arXiv:1602.05525 | DOI: 10.1103/PhysRevD.93.094515

Why selected. The standard signal-enhancement tool of every boosted-hadron calculation in this corpus: quoted by the LaMET helicity, transversity, deuteron-like dibaryon and kinematically-enhanced-operator papers (∼6/99).

Physics content. Ordinary spatial smearing optimizes overlap with the *rest-frame* wave function; a boosted hadron's wave function oscillates, so overlap dies like powers of P^z. Momentum smearing multiplies the smearing kernel by a plane-wave phase tuned to the target momentum,

$$\tilde{\psi}(\mathbf{x}) = \sum_{\mathbf{y}} e^{\,\mathrm{i}\mathbf{P}\cdot(\mathbf{x}-\mathbf{y})}\, K(\mathbf{x}-\mathbf{y})\, \psi(\mathbf{y}), \tag{2.4}$$

so the smeared operator tracks the moving state. Signal quality for three-point functions at $P^z \sim 1$–2 GeV improves severalfold — often the difference between a PDF extraction and no extraction.

Impact and echoes in this library. Combined with distillation in the library's HadStruc papers, and with optimized interpolator bases in Zhang et al.'s kinematically-enhanced operators paper (which reports further order-of-magnitude gains from P^2-enhanced overlaps), this technique defines the current art of high-momentum matrix elements.

Chapter 3

The Yang–Mills Gradient Flow Program

In 2009–2010 Martin Lüscher introduced a diffusion equation for gauge fields and turned it into one of the most fruitful ideas of modern lattice field theory. The flow smooths gauge fields over a radius $\sqrt{8t}$ while preserving gauge invariance, and the smoothed fields are *renormalized fields*: composite observables built from them are finite without additional renormalization. The library contains the original articles of this program; they anchor its two offshoots — precision scale setting and thermodynamics on one side, smeared operator product expansions and smeared quasi-PDFs on the other.

3.1 Smooth Wilson Loops: The Idea of a Flow Before the Flow

Infinite N phase transitions in continuum Wilson loop operators
R. Narayanan and H. Neuberger
JHEP **03** (2006) 064
arXiv:hep-th/0601210 | DOI: 10.1088/1126-6708/2006/03/064

Why selected. An original article of this library and the conceptual ancestor of gradient-flow smearing of Wilson lines: it is cited by the Monahan–Orginos and Monahan quasi-PDF papers, by the Brambilla–Wang off-lightcone paper, and by the higher-order flow papers (∼6/99).

Physics content. A large Wilson loop is not a well-defined observable in the continuum limit — its perimeter diverges. Narayanan and Neuberger studied a *smoothed* loop obtained by diffusing the contour, and showed numerically that the smoothed loop has a finite continuum limit. In the 't Hooft limit, enlarging the loop triggers a phase transition where the eigenvalue density of the holonomy closes its gap; if the transition falls in a solvable universality class (the Gross–Witten / Durhuus–Olesen / Douglas–Kazakov lineage cited in the paper), the string tension becomes analytically accessible.

Impact and echoes in this library. Beyond large-N physics, this paper established that smoothing a Wilson line is compatible with a finite continuum limit — precisely the fact on which smeared (pseudo-/quasi-) PDFs rest, since their nonlocal operators contain open Wilson lines. The library's flow-based PDF papers cite it as the origin of the idea.

3.2 Trivializing Maps: The Flow Enters Field Theory

Trivializing maps, the Wilson flow and the HMC algorithm
M. Lüscher
Commun. Math. Phys. **293**, 899–911 (2010)
arXiv:0907.5491 | DOI: 10.1007/s00220-009-0995-5

Why selected. An original article of this library, and the first appearance of the Yang–Mills gradient flow in the literature: the "Wilson flow" is named here. Cited throughout the flow corpus of the library (∼6/99).

Physics content. A trivializing map transforms the action to one whose long-distance couplings vanish, ideally rendering the measure Gaussian. Lüscher shows that integrating the infinitesimal version leads to a diffusion equation on field space,

$$\partial_s B_\mu(s,u) = E_\mu(B(s,u)), \qquad B_\mu(0) = U_\mu, \tag{3.1}$$

where E_μ is the negative $SU(3)$ gradient of the action — the flow. Exact maps would make Monte Carlo trivial; approximate ones can still be used inside HMC to damp topological autocorrelations. The paper already notes that short flow times probe the ultraviolet structure of the theory, setting the agenda for everything that follows.

Impact and echoes in this library. This is where the community learned that flowing fields is both a numerical tool and a renormalization concept. The AI-sampling papers of Chapter 8 are, in a precise sense, machine-learning searches for trivializing maps.

3.3 Properties of the Wilson Flow

Properties and uses of the Wilson flow in lattice QCD
M. Lüscher
JHEP **08** (2010) 071
arXiv:1006.4518 | DOI: 10.1007/JHEP08(2010)071

Why selected. The single most-cited paper of the entire citation network mined for this selection (∼15/99): every gradient-flow article of the library builds on it, and even LaMET and BMW papers quote it for scale setting or QED-flow interpolation. An original article of the library.

Physics content. The Wilson flow evolves link fields by the gauge-covariant heat equation

$$\partial_t B_\mu = D_\nu G_{\nu\mu} + \xi\, D_\mu\, \partial_\nu B_\nu, \qquad B_\mu(0) = A_\mu, \tag{3.2}$$

with $\xi = 0$ (Wilson) or $1/5$ (symmetric). Combining one-loop perturbation theory with numerical evidence, Lüscher establishes three facts: (i) $B_\mu(t)$ at any fixed $t > 0$ is a smooth field whose local gauge-invariant functionals have finite continuum limits — the flow is itself a renormalization; (ii) the observable $w(t) = t^2\langle E(t)\rangle$ with $E = \frac{1}{2}G^a_{\mu\nu}G^a_{\mu\nu}$ rises steeply through a crossover at $t \sim a^2$, so the scale w_0, defined by $\sqrt{8t}\, w(t)|_{w_0} = w_0$ (≈ 0.4 fm), provides a precision reference scale; (iii) topological sectors decouple dynamically once flowed beyond $\sqrt{8t} \sim a$.

Impact and echoes in this library. w_0/w_a scale setting is standard practice across the library's ensembles (BMW, CLQCD clover sets). More profoundly, the theorem "flowed products are UV finite" is the license for every application in this chapter and the smeared-operator applications of Chapter 6.

3.4 Renormalizability of the Flowed Theory

Perturbative analysis of the gradient flow in non-abelian gauge theories
M. Lüscher and P. Weisz
JHEP **02** (2011) 051
arXiv:1101.0963 | DOI: 10.1007/JHEP02(2011)051

Why selected. Second-highest hub of the flow corpus (∼13/99): all short-flow-time calculations in the library (energy–momentum tensor at NNLO, quark bilinears through NNLO, dipole matching) cite it as their axiomatic base. Original article of the library.

Physics content. The flow lives on the half-space $\mathbb{R}^4 \times [0, \infty)$ with boundary condition $B(0) = A$; Lüscher and Weisz formulate it as a five-dimensional local field theory, derive its Feynman rules, and prove by power counting plus BRST symmetry that correlation functions of flowed fields at strictly positive flow time require *no renormalization beyond that of the underlying four-dimensional theory.* The proof organizes divergences into counterterms that are themselves flows of renormalized lower-dimension operators, with coefficients fixed recursively; BRST Ward identities kill the dangerous classes. This is the all-orders theorem on which every "small-flow-time expansion" rests:

$$O_R(x) \xrightarrow{t\to 0} c_O(t,\mu)\,[O]_R(x) + \mathcal{O}(t), \tag{3.3}$$

with computable Wilson coefficients c_O and finite, unambiguous limits.

Impact and echoes in this library. Every matched observable in this book's chapters 3 and 6 — running couplings, energy–momentum tensor, chiral condensate, smeared twist-2 operators, smeared quasi-PDFs — is an instance of this expansion with coefficients computed within the five-dimensional formalism introduced here.

3.5 Fermions and Chiral Symmetry

Chiral symmetry and the Yang–Mills gradient flow
M. Lüscher
JHEP **04** (2013) 123
arXiv:1302.5246 | DOI: 10.1007/JHEP04(2013)123

Why selected. Third hub of the flow corpus (∼12/99): the fermion flow defined here is the basis of ringed fermions, smeared bilinears, the smeared-quasi-PDF program and the flow-dressed CP-violating operators all present in the library. Original article of the library.

Physics content. The flow is extended to quark fields by the covariant heat equation

$$\partial_t \chi = \Delta \chi, \qquad \Delta = D_\mu D_\mu, \qquad \chi(0) = \psi, \tag{3.4}$$

whose solution admits a representation as a five-dimensional local theory (the fermion propagates in one extra dimension). Consequences proved or constructed here: flowed bilinears such as $S(t) = \bar{\chi}\chi$ have finite continuum limits; the small-flow-time limit of $S(t)$ defines the renormalized scalar density, so the chiral condensate follows from

$$\lim_{t\to 0} Z_S^{-1}(t)\,\langle S(t)\rangle = -\langle \bar{\psi}\psi\rangle_R, \tag{3.5}$$

with no additive self-renormalization ambiguity; and O(a)-improved lattice actions can be constructed by adding the flow-induced dimension-5 term — the germ of "ringed" fermions later used to keep wave-function renormalization multiplicative in smeared quasi-PDFs.

Impact and echoes in this library. Monahan–Orginos's smeared quasi-PDFs (Chapter 6) and Shindler's gradient-flow moments (in the library) take the flowed fermion as their starting point; the NNLO bilinear-matching papers compute exactly the coefficients of this expansion.

3.6 The Energy–Momentum Tensor from the Flow

Energy–momentum tensor from the Yang–Mills gradient flow
H. Suzuki
Prog. Theor. Exp. Phys. **2013**, 083B03
arXiv:1304.0533 | DOI: 10.1093/ptep/ptt059

Why selected. Hub of the flow corpus (∼8/99), together with its companion Makino–Suzuki, PTEP **2014**, 063B02 [arXiv:1403.4772] which extends the formula to full QCD with fermions (∼9/99). Both are original articles of the library.

Physics content. Because flowed products are finite (3.2 above and the Lüscher–Weisz theorem), any properly normalized local product of flowed fields has an unambiguous small-flow-time expansion. Suzuki constructs gauge-invariant combinations $U_{\mu\nu}(t,x)$ of flowed fields whose expansion begins with the traceless stress tensor, and derives the master formula for pure YM:

$$T_{\mu\nu}(x) = \lim_{t\to 0}\Big[\mathcal{C}_1(t)\,U^{(1)}_{\mu\nu}(t,x) + \mathcal{C}_2(t)\,\delta_{\mu\nu}\,U^{(2)}(t,x)\Big], \tag{3.6}$$

where the coefficients are fixed by matching to dimensional regularization through the trace identity $T^\mu{}_\mu = (\beta(g)/2g)\,F^2$. The result bridges lattice and continuum schemes and turns thermodynamic quantities (trace anomaly, entropy density) into flow measurements.

Impact and echoes in this library. The Makino–Suzuki extension supplies the fermion terms used by the library's thermodynamics-oriented flow papers; Harlander–Kluth–Lange (in the library) pushed the Wilson coefficients to NNLO, making the method quantitatively competitive. Conceptually, this paper is the template for *any* flowed-composite observable: define it at finite t, expand, match. Chapter 6 will apply the same template to parton physics.

Chapter 4

From Euclidean Lattices to Light Cones: The LaMET Revolution

Parton distributions are light-cone correlations; the lattice computes Euclidean correlators. For thirty years the bridge between them was limited to Mellin moments of twist-2 operators. In 2013 Xiangdong Ji proposed computing the full x-dependence instead: boost the hadron, measure an equal-time spatial correlation, and match it perturbatively onto the light-cone distribution. This chapter collects the seven papers that define the framework (LaMET) and its coordinate-space twin (pseudo-PDFs).

4.1 The Founding Proposal

Parton physics on a Euclidean lattice
Xiangdong Ji
Phys. Rev. Lett. **110**, 262002 (2013)
arXiv:1305.1539 | DOI: 10.1103/PhysRevLett.110.262002

Why selected. The single most influential paper in the entire citation network mined for this selection (∼19/25 of the theory corpus; cited by essentially every PDF/DA/TMD article of the library). It is itself an original article of this library (`Parton Physics on Euclidean Lattice.pdf`), verified to be the PRL text.

Physics content. Define the *quasi-PDF* through the equal-time, spatially separated bilocal operator with a Wilson line,

$$Q_\Gamma(z, P^z) = \langle P|\, \bar\psi(z\hat z)\, \Gamma\, W(z\hat z, 0)\, \psi(0)\, |P\rangle, \tag{4.1}$$

which is computable on the lattice exactly like any three-point function. At large P^z, light-cone dominance emerges dynamically: a one-loop analysis gives the factorization

$$Q(x, P^z, \mu) = \int_x^\infty \frac{dy}{|x|}\, K\Big(\frac{y}{x}, \frac{\mu}{P^z}\Big)\, q(y, \mu) + \mathcal{O}\left(\frac{\Lambda^2_{\rm QCD}}{(P^z)^2}\right), \tag{4.2}$$

with K perturbatively computable and universal. Ji checked at one loop that K reproduces the Altarelli–Parisi kernel in the short-distance limit, sketched extensions to gluons, GPDs, TMDs and Wigner functions, and stated the power correction as the price of the method.

Impact and echoes in this library. This is the birth certificate of a subfield. Every chapter after this one either implements, refines, renormalizes or contests formula (4.2). Within the library, the quasi-PDF numerics (Lin group), pseudo-PDF numerics (HadStruc), TMD program (LPC) and even the agentic-workflow papers all trace their lineage to this proposal.

4.2 LaMET as an Effective Field Theory

Parton physics from large-momentum effective field theory
Xiangdong Ji
Sci. China Phys. Mech. Astron. **57**, 1407–1412 (2014)
arXiv:1404.6680 | DOI: 10.1007/s11433-014-5450-3

Why selected. Second-most-cited theory hub (∼15/99): the formal statement of Large-Momentum Effective Theory as a systematic expansion, quoted by every later methodological paper of the library.

Physics content. Boosted states behave as if governed by an effective theory whose expansion parameter is $\Lambda_{\rm QCD}/P^z$: observables factorize as

$$f(P^z) = \sum_n C_n(P^z,\mu) \otimes f_n(\mu) \left(\frac{\Lambda_{\rm QCD}}{P^z}\right)^{\omega_n}, \tag{4.3}$$

where C_n are universal matching coefficients and f_n are matrix elements of a small set of quasi-operators. The paper articulates what makes the class "universal": the same coefficients apply across hadrons and processes, power corrections are dominated by known sources, and infrared divergences cancel between Q and q. It also frames the strategy that later became standard practice — extract at moderate P^z, renormalize nonperturbatively, match, then extrapolate $\Lambda/P^z \to 0$ with controlled systematics.

Impact and echoes in this library. Together with Ji 2013 this pair defines the language in which the library's LP3/HadStruc/ETMC numerical papers write their introductions. The RMP review below is the mature statement of this program.

4.3 The First Matching Coefficient

More on large-momentum effective theory approach to parton physics
Xiangdong Ji, Jian-Hui Zhang, Yong Zhao
Nucl. Phys. B **924**, 366–376 (2017)
arXiv:1706.07416

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and

One-loop matching for parton distributions: Nonsinglet case
Xiong, Ji, Zhang, Zhao

Phys. Rev. D **90**, 014051 (2014)
arXiv:1310.7471 | DOI: 10.1103/PhysRevD.90.014051

Why selected. The 2014 letter computed the first one-loop matching kernel for the nonsinglet quasi-PDF (∼12/99 citations); the 2017 companion (∼5/99) systematized the power-counting argument and defended the approach against early criticism. Both are indispensable steps from proposal to tool.

Physics content. Working in dimensional regularization, the one-loop analysis yields the explicit kernel

$$\tilde{K}\left(\frac{x}{y}\right) = \delta(x-y) + \frac{\alpha_s}{2\pi}\, k\left(\frac{x}{y}\right), \tag{4.4}$$

with k containing both the AP splitting function and finite terms that absorb the difference between equal-time and light-cone kinematics — including the characteristic cusp anomalous dimension of the Wilson line through $\ln(P^z/\mu)$. The NPB 924 paper proves that the only power divergence of the quasi operator resides in the Wilson-line self-energy (factorizable, hence removable) and argues that Ioffe-time methods obey the same large-P^z logic, unifying the two extraction routes.

Impact and echoes in this library. These kernels are what turn lattice matrix elements into PDFs in the library's quark-PDF papers (Liu et al. 2018/2020 helicity and unpolarized extractions use precisely this nonsinglet kernel).

4.4 The Pseudo-PDF Framework

Quasi-parton distribution functions, momentum distributions, and pseudo-parton distribution functions
A. V. Radyushkin
Phys. Rev. D **96**, 034025 (2017)
arXiv:1705.01488 | DOI: 10.1103/PhysRevD.96.034025

Why selected. Defining paper of the coordinate-space route (∼12/99): every HadStruc article of the library builds on it, and the ETMC and Lin groups quote it as co-foundation.

Physics content. Radyushkin recast the same physics in Ioffe-time space. With $\nu = m\sigma$ ($m = 2\pi/T$, spacing σ), define the Ioffe-time distribution

$$\mathcal{M}(\nu, z^2) = \langle P|\bar{\psi}(z)\Gamma W(z,0)\psi(0)|P\rangle, \qquad q(\nu) = \int_{-1}^{1} dx\, e^{i\nu x} q(x). \tag{4.5}$$

The lattice measures $\mathcal{M}$ at spacelike $z^2 < 0$ — the *pseudo*-ITD. Dividing out the z-independent pion form-factor-like piece defines the *reduced* ITD, free of wave-function renormalization:

$$\mathcal{R}(\nu, z^2) \equiv \frac{\mathcal{M}(\nu, z^2)}{\mathcal{M}(0, z^2)} \xrightarrow[|z^2|\to 0]{} \int_{-1}^{1} dx\, e^{i\nu x} q(x) + \mathcal{O}(z^2 m^2, \nu^2 m^2 z^2/\nu_0^2), \tag{4.6}$$

a short-distance factorization in position space, equivalent at one loop to LaMET matching but organized entirely through OPE-type reasoning descended from Balitsky–Braun.

Impact and echoes in this library. The library's reduced-pseudo-ITD extractions (Fan–Lin pion gluon, Khan et al., Egerer et al., Salas-Chavira et al.) implement exactly this ratio; the equivalence with quasi-PDFs established by Izubuchi et al. (below) makes the two routes two dialects of one theory.

4.5 First Lattice Implementation

Lattice QCD exploration of parton pseudo-distribution functions
K. Orginos, A. Radyushkin, J. Karpie, S. Zafeiropoulos
Phys. Rev. D **96**, 094503 (2017)
arXiv:1706.05373 | DOI: 10.1103/PhysRevD.96.094503

Why selected. The first numerical realization of the pseudo-PDF idea (∼12/99): the reference point that every subsequent HadStruc and comparison plot in the library quotes.

Physics content. On quenched and $N_f = 2$ anisotropic ensembles the paper extracts the reduced pseudo-ITD from boosted pion and nucleon matrix elements, demonstrates the expected scaling with ν and z, performs the one-loop matching into x-space, and shows consistency with global-fit PDFs within uncertainties — establishing feasibility, error budgeting (excited states, z^2 truncation, momentum reach) and the workflow used ever since.

Impact and echoes in this library. Together with Radyushkin 2017 this paper founded the second numerical school of lattice parton physics. Its descendants in this library include the nucleon/pion/kaon gluon pseudo-ITD series and the polarized explorations.

4.6 The Factorization Theorem

Factorization theorem relating Euclidean and light-cone parton distributions
T. Izubuchi, X. Ji, L. Jin, I. W. Stewart, Y. Zhao
Phys. Rev. D **98**, 056004 (2018)
arXiv:1801.03917 | DOI: 10.1103/PhysRevD.98.056004

Why selected. The rigorous backbone of both routes (∼14/99): quoted whenever a library paper needs to justify that its Euclidean observable actually converges to the light-cone PDF.

Physics content. Using the operator product expansion in position space, the paper proves that the quasi-PDF obeys the LaMET factorization formula with calculable coefficients, and that the pseudo-PDF satisfies the equivalent reduced-ITD factorization; the two formulations are related by Fourier transformation order by order. Two structural results matter for practice: the matching coefficient depends on the *parton* momentum xP^z, not on the hadron momentum alone — later formalized as the natural scale choice $\mu \sim \sqrt{|x|}P^z$ and exploited by resummation papers in the library — and the proof exhibits where the power corrections live, enabling systematic error estimates.

Impact and echoes in this library. After this theorem, "quasi vs. pseudo" ceased to be competing claims and became cross-checks. The library's comparison plots between HadStruc (pseudo) and LP3 (quasi) results rest on this common ground.

4.7 The Mature Synthesis

Large-momentum effective theory
X. Ji, Y.-S. Liu, Y. Liu, J.-H. Zhang, Y. Zhao
Rev. Mod. Phys. **93**, 035005 (2021)
arXiv:2004.03543 | DOI: 10.1103/RevModPhys.93.035005

Why selected. The authoritative review (∼11/99 and rising): the canonical citation for the whole framework, written when the field's foundations had hardened.

Physics content. A comprehensive synthesis: quasi-distributions for PDFs, GPDs, TMDs and distribution amplitudes; the complete renormalization story (linear divergences, auxiliary fields, hybrid schemes); power corrections and renormalons; resummation of longitudinal logarithms; and the status of numerical results circa 2020, including the library's own LP3 program. It also clarifies conceptual points that individual letters could only assert: universality classes of quasi-observables, the role of the Wilson-line cusp, and why moments-based and x-dependent methods are complementary rather than rival.

Impact and echoes in this library. In this library the review functions as the shared textbook: the transversity and Boer–Mulders papers cite it as their definitional reference, and the agentic-workflow papers adopt its pipeline as the domain logic they automate. Reading it first makes every entry of Chapters 5–6 easier.

Chapter 5

The Renormalization Battle

The Wilson line of eq. (4.1) is a self-energy divergent object: beyond the usual logarithms it carries a *linear* UV divergence, proportional to its length z. Whether and how this can be tamed decided the fate of lattice parton physics. This chapter follows the battle in seven papers: from the veteran renormalization scheme that made lattice matrix elements respectable, through the proofs of renormalizability, to the practical schemes (ratio/RI/MOM hybrids, auxiliary fields, self-renormalization) in daily use inside this library.

5.1 RI/MOM: The Veteran Scheme

A general method for nonperturbative renormalization of lattice operators
G. Martinelli, C. Pittori, C. T. Sachrajda, M. Testa, A. Vladikas
Nucl. Phys. B **445**, 81–108 (1995)
arXiv:hep-lat/9411010 | DOI: 10.1016/0550-3213(95)00126-D

Why selected. The founding scheme of nonperturbative operator renormalization (∼8 explicit citations plus universal implicit use): every RI/MOM or SMOM renormalized matrix element in the library descends from it.

Physics content. Renormalize an operator O by imposing a renormalization condition on its matrix element in a family of off-shell Landau-gauge momentum-subtraction states,

$$Z_O(\mu, a)\,\Lambda_O(p)\big|_{p^2=\mu^2} = \Lambda_O^{tree}(p), \qquad p^2 \gg \Lambda^2_{\rm QCD}, \tag{5.1}$$

computed in lattice perturbation-free simulations. The scheme is renormalization-group closed with computable conversion factors to $\overline{\rm MS}$, and — crucially for later chapters — extends verbatim to bilocal operators, where its momentum-space amputated vertex became the diagnostic instrument for the linear divergence of Wilson lines.

Impact and echoes in this library. All early quasi-PDF extractions in the library (Liu et al., helicity at physical mass, isovector systematic study) renormalize RI/MOM-style; the TMD papers document where it fails (Chapter 6 context), which is itself part of this paper's legacy: you can only diagnose a scheme's limits because the scheme exists.

5.2 Renormalizability Proved

Renormalizability of quasiparton distribution functions
T. Ishikawa, Y.-Q. Ma, J.-W. Qiu, S. Yoshida
Phys. Rev. D **96**, 094019 (2017)
arXiv:1707.03107 | DOI: 10.1103/PhysRevD.96.094019

Why selected. First general proof that quasi-PDF operators are renormalizable (∼13/99): the license without which the program might have been a numerical artifact.

Physics content. Analyzing all one-loop diagrams of the bilocal operator, the paper shows that every power divergence localizes in the cusp-less part of the Wilson-line self-energy — i.e. the operator renormalizes multiplicatively up to mixing among operators of equal dimension with same quantum numbers, and no new nonlocal counterterm is required. Combined with a gauge-invariant UV regulator argument, the result guarantees that a continuum limit of suitably renormalized quasi-PDFs exists. The companion PRL by Ma–Qiu (arXiv:1404.6860 lineage; "good lattice cross sections") framed the same logic as a general strategy: compute at short distances where the lattice is reliable, match to the long-distance observable.

Impact and echoes in this library. Every renormalization proposal in this library — ratio schemes, auxiliary-field subtraction, hybrid schemes — cites this proof as its theoretical warrant.

5.3 Taming the Linear Divergence

Improved quasi parton distribution through Wilson line renormalization
J.-W. Chen, X. Ji, J.-H. Zhang
Nucl. Phys. B **915**, 1–9 (2017)
arXiv:1609.08102 | DOI: 10.1016/j.nuclphysb.2016.12.011

Why selected. Origin of the auxiliary-field (mass counterterm) cure for the linear divergence (∼11/99): cited by essentially every renormalization-oriented article in the corpus.

Physics content. Introduce an auxiliary field ϕ whose quadratic term carries a mass m_0; the open Wilson line re-expresses as expectation values in an enlarged theory where the linear divergence appears solely as the self-energy of ϕ, absorbed by

$$m_0^2(z,a) = m^2 + \frac{c_1}{a} + \cdots . \tag{5.2}$$

After subtraction the improved quasi-distribution contains only logarithmic divergences at any order in perturbation theory, and admits standard nonperturbative renormalization. The paper also computed the first lattice-regularized one-loop matching kernel, exposing how discretization artifacts enter K.

Impact and echoes in this library. This construction is the seed of Green–Jansen–Steffens' implementation and of the hybrid schemes below; conceptually it converts "subtracting a linear divergence" from a numerics problem into a mass renormalization problem — solvable by fitting $m_0(z)$ across z.

5.4 The Renormalization Program Formalized

Renormalization in large momentum effective theory of parton physics
X. Ji, J.-H. Zhang, Y. Zhao
Phys. Rev. Lett. **120**, 112001 (2018)
arXiv:1706.08962 | DOI: 10.1103/PhysRevLett.120.112001

Why selected. The programmatic statement of LaMET renormalization (∼13/99): the reference for auxiliary-field formalism and for the Wilson-line anomalous dimension entering matching at NNLO.

Physics content. The letter systematizes the two-step logic now used everywhere: (i) renormalize the quasi-operator nonperturbatively within a chosen scheme (RI/MOM, ratio, or auxiliary-field mass), rendering the matrix element finite in the continuum limit; (ii) convert to $\overline{\text{MS}}$ and match to the light-cone distribution with kernels that include the cusp logarithm $\ln(P^z/\mu)$. It also computes the one-loop conversion factors and demonstrates scheme independence of the final light-cone PDF order by order.

Impact and echoes in this library. The library's hybrid-renormalization applications (transversity, Boer–Mulders, gluon PDFs) cite this letter as their definitional framework; its notation is the de facto standard across LP3 papers.

5.5 Auxiliary Fields Implemented

Nonperturbative renormalization of nonlocal quark bilinears from lattice QCD
J. Green, K. Jansen, F. Steffens
Phys. Rev. Lett. **121**, 022004 (2018)
arXiv:1707.07152 | DOI: 10.1103/PhysRevLett.121.022004

Why selected. First numerical demonstration that the auxiliary-field scheme works on realistic ensembles (∼14/99): the bridge between the Chen–Ji–Zhang idea and everyday practice.

Physics content. On $N_f = 2 + 1$ domain-wall ensembles the paper measures the quasi-PDF operator's Green function, fits the linearly divergent ϕ-field mass $m_0(z, a)$ over a range of separations and spacings, subtracts it, and shows the residual matrix element scales logarithmically toward a finite continuum limit — then supplies the conversion factors to $\overline{\text{MS}}$. The fit-based extraction of a power-divergent quantity, with its correlated systematics, became the template that self-renormalization (below) would generalize.

Impact and echoes in this library. Within the library, the ETMC and LPC lines quote this paper when justifying hybrid short-distance treatments; its error-analysis patterns reappear in every modern extraction.

5.6 The Complete Prescription

A complete non-perturbative renormalization prescription for quasi-PDFs
C. Alexandrou, K. Cichy, M. Constantinou, K. Hadjiyiannakou, K. Jansen, H. Panagopoulos, F. Steffens
Nucl. Phys. B **923**, 394–415 (2017)
arXiv:1706.00265 | DOI: 10.1016/j.nuclphysb.2017.08.010

Why selected. The most complete single-scheme treatment (∼10/99): the ETMC school's standard, shared by many library papers including the twisted-mass gluon PDF.

Physics content. For unpolarized, helicity and transversity quasi-PDFs the paper constructs the full RI′ prescription: renormalization conditions for all relevant tensor structures, subtraction of the linear divergence via the z-dependent mass term, inclusion of the mixing with the scalar operator for twist-3 structures, and perturbative conversion factors RI′ → $\overline{\text{MS}}$ at one loop. The result is a recipe that a simulation can follow end-to-end — matrix element, Z factors, matching, extrapolations — which explains its role as the field's practical benchmark.

Impact and echoes in this library. In this library it anchors the ETMC proton gluon-PDF calculation and provides the comparison point against hybrid-ratio schemes; the gauge-fixing-precision paper quantifies exactly how much care its Landau-gauge step demands.

5.7 Gluon Operators Renormalize Multiplicatively

Multiplicative renormalizability of operators defining quasiparton distributions
Z.-Y. Li, Y.-Q. Ma, J.-W. Qiu
Phys. Rev. Lett. **122**, 062002 (2019)
arXiv:1809.01836 | DOI: 10.1103/PhysRevLett.122.062002

Why selected. The rigorous basis for gluon quasi-PDFs (∼8/99): quoted by nearly every gluon-PDF article of the library.

Physics content. Classifying all gauge-invariant bilocal adjoint operators with a Wilson line, the paper proves — using a gauge-invariant UV regulator so that only genuine UV singularities appear — that the full set of 36 gluon quasi-parton operators has localized UV divergences and renormalizes *multiplicatively*: each operator maps onto a finite combination fixed by symmetry, with no operator mixing beyond the trivial ones. This completes, for gluons, what Ishikawa et al. proved for quarks, and certifies that lattice gluon quasi-PDFs define continuum quantities.

Impact and echoes in this library. Together with Zhang et al.'s matching analysis (next chapter), this theorem is why "first gluon PDF from the lattice" papers can exist at all. The library's gluon-quasi-PDF line cites it at every methodological juncture.

Chapter 6

Gluon PDFs: Where the Two Lines Converge

Gluon distributions are where the field's difficulties concentrate: adjoint operators, larger statistical noise, single-charge-conjugation mixing with quarks in the singlet channel — and the largest theory uncertainty in global fits. The five papers of this chapter supplied the theory for the library's gluon-PDF program and fused the gradient-flow and large-momentum lines.

6.1 Matching Gluon Quasi-PDFs

Accessing gluon parton distributions in large momentum effective theory
J.-H. Zhang, X. Ji, A. Schäfer, W. Wang, S. Zhao
Phys. Rev. Lett. **122**, 142001 (2019)
arXiv:1808.10824 | DOI: 10.1103/PhysRevLett.122.142001

Why selected. The theoretical engine of every gluon quasi-PDF calculation in the library (∼9/99).

Physics content. Using an auxiliary adjoint heavy-quark field coupled through the Yang–Mills Lagrangian as a gauge-invariant regulator, the paper proves that all power divergences of gluon quasi-PDF operators reside in the Wilson-line self-energy and are removable by mass renormalization. It then constructs the independent multiplicatively renormalizable correlators — four unpolarized, three helicity — computes their one-loop matching to light-cone gluon PDFs, and states the LaMET factorization formula for the gluon channel. The classification also clarifies which operator combinations are free of mixing with quark singlets at leading twist.

Impact and echoes in this library. The library's Fan et al. gluon quasi-PDF letter and the Lin-group LaMET gluon series implement precisely this framework; the hybrid-renormalization papers adapt its subtraction to finite-flow operators.

6.2 Gluon Pseudo-Distributions from Short Distances

Gluon pseudo-distributions at short distances: forward case
I. Balitsky, W. Morris, A. Radyushkin

Phys. Lett. B **808**, 135621 (2020)
arXiv:1910.13963 | DOI: 10.1016/j.physletb.2020.135621

Why selected. Theory hub of the entire HadStruc gluon line (∼10/99): every gluon pseudo-ITD extraction in this library quotes its matching kernel.

Physics content. The paper classifies bilocal double-gluon operators and isolates the twist-2 invariant amplitudes entering forward matrix elements; using the background-field method and heat-kernel techniques it computes the short-distance expansion of the gluon pseudo-ITD in an explicitly gauge-invariant form,

$$\mathcal{M}_g(\nu, z^2)\big|_{|z^2|\ \text{small}} = C_g(\nu, z^2, \mu) \otimes g(x, \mu) + \mathcal{O}(z^2), \tag{6.1}$$

separating ultraviolet logarithms from DGLAP evolution and supplying the reduced-ITD matching relation. The companion paper (PRD **105**, 014008 (2022), arXiv:2111.06797, also cited widely in the library) provides complete expressions for all projections and the quark-mixing terms.

Impact and echoes in this library. Khan et al., Egerer et al., Salas-Chavira et al., Fan–Lin and the physical-point/continuum extractions all convert their matrix elements through these kernels — they are to gluon pseudo-PDFs what Xiong et al. is to the nonsinglet quark channel.

6.3 Smeared Quasi-PDFs: The Flow Meets Partons

Quasi parton distributions and the gradient flow
C. Monahan, K. Orginos
JHEP **03** (2017) 116
arXiv:1612.01584 | DOI: 10.1007/JHEP03(2017)116

Why selected. Original article of this library and the explicit bridge between Chapters 3 and 4 (∼5/99 within its niche, universal within the smeared-operator subline): it imports the flow's finiteness theorem into parton physics.

Physics content. Replace the bare Wilson-line operator of eq. (4.1) by a version built from flowed fields $B_\mu(t)$, $\chi(t)$ at small flow time t: since flowed products are UV finite (eq. (3.2)), the matrix element has a smooth continuum limit without manual subtraction of linear divergences. With ringed fermions the wave-function renormalization stays multiplicative; the small-t expansion relates moments of the smeared quasi-PDF to light-cone moments with computable coefficients obeying a DGLAP-like system. The follow-up one-loop analysis by Monahan (PRD **97**, 054507 (2018), in the library) supplies the matching kernels and confirms that smearing leaves infrared structure untouched.

Impact and echoes in this library. This is the conceptual source of the library's "Gradient Flow for PDFs" pion application and of Shindler's flowed twist-2 moment proposal: once operators are smeared by the flow, the renormalization battle of Chapter 5 simplifies into ordinary matching — at the price of a new scale t whose systematic treatment the NNLO bilinear papers provide.

6.4 Self-Renormalization

Self-renormalization of quasi-light-front correlators on the lattice
Y.-K. Huo et al. (Lattice Parton Collaboration)
Nucl. Phys. B **969**, 115443 (2021)
arXiv:2103.02965 | DOI: 10.1016/j.nuclphysb.2021.115443

Why selected. Original article of this library and the workhorse scheme of its LP3 program (∼6/99): used by the transversity, DA and TMD papers.

Physics content. Exploit the structure of the divergence itself: write the renormalized matrix element as

$$Q_R(z, P^z; \mu) = Z^{-1}(z, a, m_0)\, Q_{\rm lat}(z, P^z, a), \qquad Z(z, a) \propto e^{-c\, z/a} \times (\text{logs}), \tag{6.2}$$

and determine both c and the residual log factor by fitting multi-spacing data on $Q_{\rm lat}$ itself — no external vertex calculation, no Landau-gauge dependence, no auxiliary field. The paper verifies universality of the extracted Z across hadron momenta and channels, and documents sizable nonperturbative contamination in RI/MOM-style vertices at long distances, motivating short-distance-only usage.

Impact and echoes in this library. Within the library this is the scheme behind several flagship results (nucleon transversity in the continuum limit; pion/kaon DAs; TMD wave functions); its diagnostics fed directly into the hybrid-scheme design below.

6.5 Hybrid Renormalization

Hybrid renormalization scheme for quasi light-front correlations in large-momentum effective theory
X. Ji, Y.-S. Liu, A. Schäfer, W. Wang, Y.-B. Yang, J.-H. Zhang, Y. Zhao
Nucl. Phys. B **964**, 115311 (2021)
arXiv:2008.03886 | DOI: 10.1016/j.nuclphysb.2021.115311

Why selected. Original article of this library; the current standard scheme of its LP3 program (∼10/99): the pragmatic synthesis after a decade of battle.

Physics content. Split the separation axis into regimes. For $z < z_c$ use a ratio of matrix elements (or RI/MOM) at short distance, where perturbation theory is trustworthy and nonperturbative pollution is absent; for $z > z_c$, subtract the linearly divergent Wilson-line self-energy through an auxiliary-field mass term matched continuously onto the short-distance regime, then convert to $\overline{\text{MS}}$. The scheme carries one parameter m_0^2 (equivalently z_c) with intrinsic $\mathcal{O}(a)$ ambiguity, for which the paper prescribes determination procedures — fitting across spacings, or fixing via the cusp anomalous dimension. Combined with momentum-space resummation of $\ln(p_z/\mu)$ logs (Su et al., NPB **991**, 116201 (2023), arXiv:2209.01236, in the library) and renormalon-consistent power corrections (Zhang et al., PLB **844**, 138081 (2023)), it defines the modern pipeline.

Impact and echoes in this library. Every late-stage LP3 result in this library — gluon PDFs, Boer–Mulders, heavy-meson LCDAs, TMD soft functions — runs on this scheme or its self-renormalization variant; the two papers together are the most-quoted methodological pair of the corpus's second half.

Chapter 7

TMDs and Global Fitting: Contact with Phenomenology

Two interfaces connect the lattice program to the rest of particle physics. Vertically, transverse-momentum-dependent distributions extend the collinear picture and bring their own factorization physics — born in the paper below. Horizontally, global fits turn thousands of data points into the PDF sets that lattice calculations benchmark against and increasingly absorb as input. One landmark of each line is included here, plus the TMD founding document.

7.1 Back-to-Back Jets: The Birth of TMD Physics

Back-to-back jets in QCD
J. C. Collins, D. E. Soper
Nucl. Phys. B **193**, 381 (1981)
DOI: 10.1016/0550-3213(81)90506-7 | companion: Collins–Soper–Sterman, Nucl. Phys. B **250**, 199 (1985)

Why selected. The definitional paper of transverse-momentum-dependent factorization (∼4/99 explicit, universal within the library's TMD line): every quasi-TMD article builds on its concepts.

Physics content. For color-singlet production with small transverse momentum, collinear factorization fails; observables depend on a new class of matrix elements with both light-cone and transverse separations. Collins and Soper introduced the rapidity cut ζ to define these TMDs, proved their factorization from hard, jet and soft sectors, and derived the evolution equation in rapidity — the Collins–Soper kernel — supplemented by DGLAP in $k_\perp$:

$$\frac{\partial \ln F(x, \mathbf{k}_\perp; \zeta, \mu)}{\partial \ln \sqrt{\zeta}} = K(b_\perp; \mu), \qquad b_\perp \leftrightarrow \mathbf{k}_\perp. \tag{7.1}$$

The companion CSS paper completed the proof for Drell–Yan. The kernel K is intrinsically nonperturbative at large $b_\perp$ — which is precisely where the lattice enters decades later.

Impact and echoes in this library. The library's quasi-TMD wave-function papers, soft-function extractions and nonperturbative Collins–Soper-kernel determinations all target this object; their renormalization puzzles (rapidity divergences on the lattice) are direct descendants of the structure this paper discovered.

7.2 CT18: The Hessian Standard

New CTEQ global analysis of quantum chromodynamics with high-precision data from the LHC
T.-J. Hou et al. (CTEQ-TEA Collaboration)
Phys. Rev. D **103**, 014013 (2021)
arXiv:1912.10053 | DOI: 10.1103/PhysRevD.103.014013

Why selected. Original article of this library and its most-quoted phenomenological benchmark (∼6/99 among fit-corpus citations): the default comparison set for nearly every lattice PDF extraction in the corpus.

Physics content. CT18 evolves the CTEQ methodology to the full LHC era: Hessian error analysis over an extended basis, NNLO theory with heavy-quark scheme variants, and careful treatment of dataset tensions (notably the ATLAS 7 TeV W/Z measurements, accommodated in the CT18A/CT18Z variants). It delivers the standard-candle predictions (Higgs channels, W mass sensitivity) against which both collider analyses and first-principles inputs are judged. Its predecessor CT14 (Dulat et al., PRD **93**, 033006 (2016), also an original article of this library) established the parameterization framework; the Hessian method itself goes back to Pumplin et al., JHEP **07**, 012 (2002).

Impact and echoes in this library. In this library CT18 appears as the reference curve in gluon-PDF, quasi-PDF and DA comparisons; the "lattice PDFs enter the global fit" program is calibrated by asking how much CT-class uncertainties shrink when lattice matrix elements join the data pool.

7.3 NNPDF3.1: The Monte Carlo School

Parton distributions from high-precision collider data
R. D. Ball et al. (NNPDF Collaboration)
Eur. Phys. J. C **77**, 663 (2017)
arXiv:1706.00428 | DOI: 10.1140/epjc/s10052-017-5199-5

Why selected. The Monte-Carlo-methodology counterpart of CT18 (∼8/99), original article of this library; its successor NNPDF4.0 (*The Path to Proton Structure at One-Percent Accuracy*) is likewise in the library and cites it as baseline.

Physics content. NNPDF represents PDFs as neural-network ensembles fitted by Monte Carlo, with closure tests certifying that uncertainties neither underestimate nor inflate truth. Version 3.1 added independent charm (theory-motivated but fitted), top-differential and high-precision LHC electroweak-boson data, reaching the accuracy regime where theory errors (NNLO splitting functions, electroweak corrections) dominate. The methodological chain — closure tests, replica compression, positivity and sum-rule enforcement — became the field's quality standard and, through NNPDF4.0's hyperparameter optimization, converged toward machine-learning engineering itself.

Impact and echoes in this library. Together the two fitting schools bracket every lattice result of this corpus: plots throughout the library overlay extractions on CT18 and NNPDF bands simultaneously. The deeper convergence — global fitting as machine learning, lattice sampling as generative modeling — is the subject of the next chapter.

Chapter 8

Stochastic Quantization Reborn: AI Sampling

The newest pillar of the library is machine-learning sampling. Its physical root is forty years older: stochastic quantization represents quantum field theory as Langevin diffusion, and generative diffusion models are exactly time-reversed Langevin dynamics. Two original articles of the library founded the lattice application.

8.1 Stochastic Quantization

Perturbation theory without gauge fixing
G. Parisi, Y.-S. Wu
Sci. Sinica **24**, 483 (1981)
No arXiv (preprint era)

Why selected. Root of the entire AI-sampling line (∼3/99 within the AI corpus, plus conceptual citations in flow papers): the idea that a quantum field theory equals the equilibrium statistics of a stochastic process predates, and now powers, the machine-learning era.

Physics content. Promote fields to functions of a fictitious fifth time s governed by Langevin dynamics,

$$\frac{\partial A_\mu(x,s)}{\partial s} = -\frac{\delta S[A]}{\delta A_\mu(x,s)} + \eta_\mu(x,s), \qquad \langle \eta_\mu(x,s)\eta_\nu(y,s')\rangle = 2\delta_{\mu\nu}\delta(x-y)\delta(s-s'), \tag{8.1}$$

whose equilibrium distribution is e^{-S}. Because the drift is gauge covariant, no gauge fixing is needed — perturbation theory proceeds in the physical (transverse) subspace automatically, and the Fokker–Planck operator generates the renormalization group (Zinn–Justin). The flow-time structure will reappear verbatim in Lüscher's gradient flow, with deterministic drift replacing the noise average.

Impact and echoes in this library. Wang–Aarts–Zhou's diffusion-as-stochastic-quantization paper (in the library) makes the correspondence exact and constructive: training a score-based diffusion model reproduces the reverse Langevin process, yielding Boltzmann-distributed samples of the field theory.

8.2 Flow-Based MCMC

Flow-based generative models for Markov chain Monte Carlo in lattice field theory
M. S. Albergo, G. Kanwar, P. E. Shanahan
Phys. Rev. D **100**, 034515 (2019)
arXiv:1904.12072 | DOI: 10.1103/PhysRevD.100.034515

Why selected. Founding article of machine-learning sampling for lattice field theory, and an original article of this library (∼4/99 within its niche): cited by every AI-for-lattice paper of the corpus.

Physics content. Train an invertible neural network f_ϕ as a proposal distribution $q_\phi(\phi) = p_0(f_\phi(\phi))|\det J_{f_\phi}|$, then use it inside Metropolis–Hastings with acceptance $\min(1, e^{-S[\phi]}/e^{-S[f_\phi(\phi)]}\cdot q/p)$: exactness of the target distribution survives any imperfection of the learned model. Demonstrated on 2d ϕ^4 theory, the approach suppresses critical slowing-down precisely where local algorithms fail, and trains without samples of the target distribution — only the action enters the loss. This resolves the classic chicken-and-egg problem of simulation-based learning.

Impact and echoes in this library. Within the library this paper is the common ancestor of the equivariant-gauge extension below, of the diffusion-as-stochastic- quantization line, and of the physics-driven-learning agenda for inverse problems; conceptually it realizes the trivializing-map dream of Chapter 3 with learnable maps.

8.3 Gauge Equivariance at Scale

Equivariant flow-based sampling for lattice gauge theory
G. Kanwar, M. S. Albergo, D. Boyda, K. Cranmer, D. C. Hackett, S. Racanière, D. J. Rezende, P. E. Shanahan
Phys. Rev. Lett. **125**, 121601 (2020)
arXiv:2003.06413 | DOI: 10.1103/PhysRevLett.125.121601

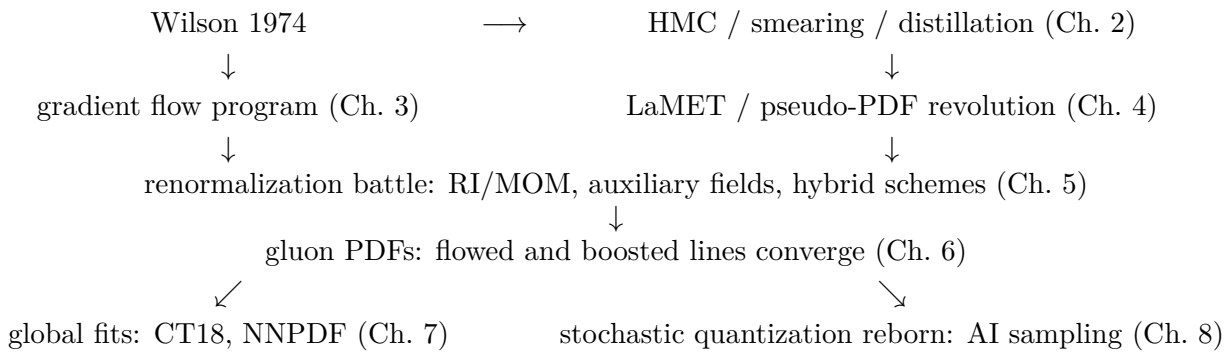
Why selected. Original article of this library; the demonstration that machine learning can attack the hardest sampling problem of lattice QCD — topology (∼3/99 within its niche).

Physics content. Build the flow from layers that commute with gauge transformations (maskings coupling links to plaquette-derived functions, coupling kernels projected onto the group through exponentiated traces), so that $q_\phi(U^g) = q_\phi(U)$ and the sampled ensemble is manifestly gauge invariant. Applied to 2d $U(1)$ pure gauge near the continuum limit, trained models reduce the autocorrelation time of the topological charge by factors of ~ 1500 relative to HMC and ~ 200 relative to heat bath — attacking the exponential freezing of topology that plagues fine-lattice QCD simulations cited since Lüscher's own work on open boundary conditions.

Impact and echoes in this library. The library frames its AI-for-QCD narrative around this result: symmetry-respecting architectures are not a constraint but the source of the gain — the same lesson its agentic-computing papers draw when they insist that deterministic domain tools must remain in charge of the physics while language models orchestrate. The circle back to Parisi–Wu and to trivializing maps closes here.

Epilogue: One Diagram of the Field

Looking back at the 38 entries, the intellectual architecture can be drawn as one diagram:



Three observations close this selection.

First, the field's decisive theoretical events were cheap in machinery but expensive in thought. The quasi-PDF proposal is a one-page definition; the factorization theorem is an OPE argument; the flow's finiteness is a power-counting proof. What made them decisive was that each converted a conceptual impossibility — "the lattice cannot reach light cone x-dependence", "composite operators need infinite counterterms" — into a finite, computable program.

Second, convergence beats competition. Quasi- and pseudo-PDFs, flowed and boosted operators, Hessian and Monte Carlo fits: every pair of rivals turned out to be two coordinates on one object, and the cross-checks between them are now the field's main systematics tool.

Third, this corpus is not closed. Its newest layer — agentic workflows that orchestrate the deterministic pipeline documented in these chapters — treats the papers above as domain logic to be automated rather than replaced. Whether that layer earns a chapter of its own in a future edition depends on precisely the standards this selection was compiled under: reproducibility, error control, and physics first.